

Tong Statistical Field Theory — Sheet 1: Landau Theory

D. Tong, Statistical Field Theory, Example Sheet 1; 4 worked problems.

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These prompts paraphrase David Tong's Statistical Field Theory, Example Sheet 1 (October 2018). Problem numbers, starred status, equations, and guidance follow the official sheet.

1. Sheet 1, Problem 1: Transfer Matrix for the One-Dimensional Ising Model

Take N Ising spins with $s_{N+1} = s_1$ and partition function

$$Z = \sum_{s_1=\pm 1} \cdots \sum_{s_N=\pm 1} \prod_{i=1}^N \exp \left[\beta J s_i s_{i+1} + \frac{\beta B}{2} (s_i + s_{i+1}) \right]. \quad (1.1)$$

Show that $Z = \text{tr } T^N$ for the transfer matrix

$$T = \begin{pmatrix} e^{\beta J - \beta B} & e^{-\beta J} \\ e^{-\beta J} & e^{\beta J + \beta B} \end{pmatrix}. \quad (1.2)$$

Hint: Express the i th factor as $T_{s_i, s_{i+1}} = \exp[\beta J s_i s_{i+1} + \beta B(s_i + s_{i+1})/2]$, with s_i labelling a column and s_{i+1} a row. Find the two eigenvalues,

$$\lambda_{\pm} = e^{\beta J} \cosh(\beta B) \pm \sqrt{e^{2\beta J} \cosh^2(\beta B) - 2 \sinh(2\beta J)}, \quad (1.3)$$

and explain why $Z \simeq \lambda_+^N$ as $N \rightarrow \infty$. At $B = 0$, deduce that the magnetisation is always zero and that temperature causes no phase transition in this one-dimensional system.

Solution. The partition function can be written as

$$Z = \sum_{s_1=\pm 1} \cdots \sum_{s_N=\pm 1} T_{s_1, s_2} T_{s_2, s_3} \cdots T_{s_N, s_1} = \text{tr } T^N = \lambda_+^N + \lambda_-^N. \quad (1.4)$$

Clearly, $\lambda_+ > \lambda_-$. Therefore, for Large N , we can write $Z \simeq \lambda_+^N$. The magnetisation is given by

$$M = \frac{1}{\beta} \frac{\partial \log Z}{\partial B} = \frac{1}{\beta Z} \left[N \lambda_+^{N-1} \frac{\partial \lambda_+}{\partial B} + N \lambda_-^{N-1} \frac{\partial \lambda_-}{\partial B} \right]. \quad (1.5)$$

Clearly, $\left. \frac{\partial \lambda_{\pm}}{\partial B} \right|_{B=0} = 0$, which gives $M \Big|_{B=0} = 0$. Therefore, there is no second-order phase transition for a one-dimensional system.

2. Sheet 1, Problem 3: Exact Mean Field Limit of a Long-Range Ising Model

Let every spin interact with every other spin, including itself, through

$$E = -B \sum_i s_i - \frac{J}{2N} \left(\sum_i s_i \right) \left(\sum_j s_j \right). \quad (2.1)$$

Explain the factor $1/N$. Establish the Gaussian identity

$$e^{\beta J \alpha^2 / (2N)} = \sqrt{\frac{N\beta J}{2\pi}} \int_{-\infty}^{\infty} dx e^{-N\beta J x^2 / 2 + \alpha\beta J x}, \quad (2.2)$$

and use it to express the partition function as

$$Z = \sqrt{\frac{N\beta J}{2\pi}} \int_{-\infty}^{\infty} dx e^{-NS(x)}, \quad S(x) = \frac{\beta J x^2}{2} - \log[2 \cosh(\beta(B + Jx))]. \quad (2.3)$$

Find the saddle x_* that dominates at large N , up to its prefactor. Derive its stationarity equation and show that $m = x_*$.

Solution. Since we expect the energy to scale as N , while $\sum_i s_i$ also scales as N . Therefore, we must have a factor of $1/N$.

The Gaussian identity is straightforward. Let us define

$$\sum_i s_i \equiv \alpha. \quad (2.4)$$

Then,

$$\begin{aligned} Z &= \sum_{\alpha} e^{\beta J \alpha^2 / (2N) + \beta B \alpha} \\ &= \sqrt{\frac{N\beta J}{2\pi}} \sum_{\alpha} e^{\beta B \alpha} \int_{-\infty}^{\infty} dx e^{-N\beta J x^2 / 2 + \alpha\beta J x} \\ &= \sqrt{\frac{N\beta J}{2\pi}} \int_{-\infty}^{\infty} dx e^{-N\beta J x^2 / 2} \sum_{\alpha} e^{\beta(B + Jx)\alpha} \\ &= \sqrt{\frac{N\beta J}{2\pi}} \int_{-\infty}^{\infty} dx e^{-N\beta J x^2 / 2} \cdot 2^N \cosh^N \beta(B + Jx) \\ &= \sqrt{\frac{N\beta J}{2\pi}} \int_{-\infty}^{\infty} dx e^{-NS(x)}. \end{aligned} \quad (2.5)$$

The saddle is given by

$$S'(x_*) = 0 \implies x_* = \tanh(\beta B + \beta J x_*), \quad (2.6)$$

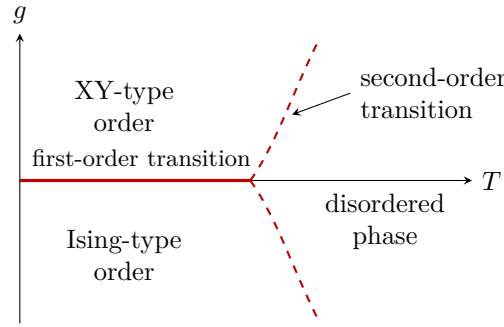
which satisfies the same equation as m as we have seen. Therefore, we have $m = x_*$, which means that the long-range Ising model realises the mean field theory exactly.

3. Sheet 1, Problem 6: Ising and XY Orders in an Anisotropic Spin Model

At each lattice site put a unit three-vector $\mathbf{s}_i = (s_i^x, s_i^y, s_i^z)$ and use

$$E = -J \sum_{\langle ij \rangle} \mathbf{s}_i \cdot \mathbf{s}_j + g \sum_i \left[(s_i^z)^2 - \frac{(s_i^x)^2 + (s_i^y)^2}{2} \right], \quad J > 0. \quad (3.1)$$

Use mean field reasoning to account for the phase diagram below,



Hint: Compare the ground states as g crosses zero at low T , then raise T separately on either side of $g = 0$. Detailed path-integral calculations are unnecessary.

Solution. The energy can be rewritten as

$$E = -J \sum_{\langle ij \rangle} \mathbf{s}_i \cdot \mathbf{s}_j + g \sum_i \left[\frac{3}{2} (s_i^z)^2 - \frac{1}{2} \right]. \quad (3.2)$$

When T is small and $g > 0$, to minimise the total energy, the system tends to align while the second term forces them into the $x - y$ plane. Therefore, it is a XY-type order. For $g < 0$, energy is minimised when all of the spins are in the z -direction and aligned. Hence, it is a Ising-type ordering. Therefore, going from $g < 0$ to $g > 0$ is a first-order phase transition.

As the temperature goes up, the system becomes disordered for both $g < 0$ and $g > 0$. Therefore, a second-order phase transition must have occurred.

4. Sheet 1, Problem 8: Multicritical Scaling and Upper Critical Dimension

Take

$$f(m) = \alpha_2 m^2 + \alpha_{2n} m^{2n}, \quad n \in \mathbb{Z}_+. \quad (4.1)$$

Find β from $m \sim (T_c - T)^\beta$ in the ordered phase. Insert the result into the free energy to show that the heat-capacity exponent obeys $\alpha = 1 - 2\beta$. Compare this with the contribution to the free energy from fluctuations in the Gaussian path integral and establish the threshold $d_c = 2n/(n - 1)$ above which the mean field term dominates near criticality.

Solution. It can be easily obtained that $\beta = 1/(2n - 2) < 1$. Plugging back into the free energy and noting that $1 + 2\beta = 2n\beta$, we have

$$f \sim \alpha_2^{1+2\beta} \implies c \sim \alpha_2^{2\beta-1}, \quad (4.2)$$

and hence $\alpha = 1 - 2\beta$. In the Gaussian path integral, we have

$$c \sim |T - T_c|^{-2+d/2}. \quad (4.3)$$

Therefore, the critical dimension is

$$d_c = \frac{2n}{n-1}. \quad (4.4)$$